
SMART STUDENT PRODUCTIVITY PLATFORM

Assignment 2: User Roles, Access Control and Authentication

Project	Smart Student Productivity Platform
Source	Software Requirements Specification (SRS), Version 1.1
Organization	SR Pvt Ltd
Status	Draft
Prepared By	Project Team
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1. Objective

- Understand the four primary user roles.
- Explain role-based access and academic scope.
- Document the Firebase Authentication flow.
- Explain authorization, RBAC, account provisioning and deactivation.

2. User Hierarchy

The platform follows a hierarchical access model so users can access only functionality and academic information relevant to their responsibilities. The SRS defines four primary roles.

- Super Admin — Platform Level
- Domain Admin / HOD — Department Level
- Faculty — Academic Management
- Student — Academic Activities

3. Role Access Matrix — Key Areas

The SRS defines different access levels for Student, Faculty, HOD and Super Admin. Access varies by module and is limited according to role and permitted scope.

- Dashboard: role-specific view
- Profile: manage own profile
- Courses and learning materials: role-specific access
- Assignments and assessments: students view/submit, faculty create/evaluate, HOD monitors, Super Admin manages
- Attendance and results: role-specific access
- User Management and Academic Structure: HOD domain scope; Super Admin platform scope
- Reports: HOD department-level; Super Admin platform-level
- Audit Logs: authorized HOD access; full Super Admin access

4. Authentication Overview

Authentication verifies the identity of a user before protected functionality can be accessed. The platform shall use Firebase Authentication.

The SRS flow is: User → Login Page → Enter Credentials → Firebase Authentication. Invalid credentials produce an error. Valid authentication leads to role retrieval, academic-scope checking and redirection to the authorized portal.

5. Authorization

Authorization determines what an authenticated user is permitted to access. The SRS states that authorization shall consider the user's role and academic scope.

- User role
- Domain/department
- Program
- Semester
- Section
- Assigned subjects
- User account status

6. Role-Based Access Control and Account Management

The system shall enforce Role-Based Access Control (RBAC). An authenticated user is identified as Student, Faculty, HOD or Super Admin and is directed to the corresponding portal.

User accounts shall be created or provisioned by authorized administrators; unrestricted public self-registration is not provided.

When an account is deactivated, the user cannot authenticate, institutional records may be retained according to retention requirements, and access permissions are no longer available.

7. Conclusion

This assignment documents the access-control flow from authentication through role identification and academic-scope verification, connecting each authenticated user to the appropriate authorized portal.